$Fibonacci\ Series\ -\ Growth\ Rate\ of\ Output$

vs

$Time\ Complexity$

$Here\ we\ have$:

```
    return fibonacci(n - 1) + fibonacci(n - 2);
```

$Time\ Complexity: T(n) = T(n-1) + T(n-2) + O(1)$

$Growth\ Rate\ of\ Output: F(n) = F(n-1) + F(n-2)$

$1)\ First\ Method\ \big(Assumming\ F(n-1) \approx F(n-2)\big)\ \begin{bmatrix} Not \\ Precise \end{bmatrix}$

$From\ time\ complexity\ we\ came\ to\ know\ that$:

$$T(n) = \begin{cases} 1 & if(n = 0) \\ 1 & if(n = 1) \\ T(n-1) + T(n-2) & if(n > 1) \end{cases}$$

$We\ know\ , Fib(0) = 0\ and\ Fib(1) = 1\ hence\ both\ are\ constants\ i.e.`C`\ ,$

$we\ can\ use\ `1\ or\ C`\ just\ to\ represent\ constant. Not\ from\ Time\ Complexity\ but$

$like\ Time\ Complexity\ .$

$F(n) = F(n-1) + F(n-2)$

$Assumming\ F(n-1) \approx F(n-2)\ Then$:

$F(n) = F(n-1) + F(n-1)$

$$F(n) = 2F(n-1)$$

$$Hence\ , for\ F(n-1) = 2 \times 2F(n-2) = 4F(n-2) = 2^2F(n-2)$$

$$Therefore\ , substituting\ , we\ get:$$

$$F(n) = 2^2F(n-2)$$

$$By\ doing\ this\ upto\ `i`\ times\ we\ get:$$

$$F(n) = 2^iF(n-i)$$

$$If\ , n-i = 1\ , then : i = n-1\ , replacing\ `i`\ , we\ get:$$

$$F(n) = 2^{n-1}F\big(n-(n-1)\big)$$

$$F(n) = 2^{n-1}F(1)$$

$$F(n) = 2^{n-1} \times 1\ [As, F(1) = 1\ or\ C]$$

$$F(n) = 2^{n-1}$$

$$We\ can\ also\ write\ it\ as :$$

$$F(n) = 2^n \times 2^{-1}$$

$$F(n) = 2^n \times \frac{1}{2}$$

$$Applying\ theta\ `\Theta`\ \text{we get:}$$

$$F(n) = \Theta(2^n) \times \Theta\left(\frac{1}{2}\right)$$

$F(n) = \Theta(2^n)$ [*This is not precise .*]

2) *Assuming* $F(n) = x^n$

Why this guess?
Because:

- ***Recurrences often behave like exponentials***
- ***Similar to how linear differential equations use*** e^x

Therefore , ***we replace every term as***:

$$F(n) = F(n-1) + F(n-2)$$

$$x^n = x^{n-1} + x^{n-2}$$

we want to *eliminate the dependence on* `**n**` ***and get a*** *constant equation in* x.

We want to convert everything into the ***same power base level***.

So we divide by the ***smallest power present***, *which is*: x^{n-2}

Dividing both side with `$x^{n-2}$` *we get*:

$$\frac{x^n}{x^{n-2}} = \frac{x^{n-1}}{x^{n-2}} + \frac{x^{n-2}}{x^{n-2}}$$

$$\Longrightarrow x^{n-(n-2)} = x^{n-1-(n-2)} + x^{n-2-(n-2)}$$

$$\Longrightarrow x^2 = x^1 + x^0$$

$$\Longrightarrow x^2 = x + 1$$

$$\Longrightarrow x^2 - x - 1 = 0$$

$By\ Quadratic\ Formula$:

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$Where, a = 1\ , b = -1\ and\ c = -1$

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times 1 \times -1}}{2 \times 1}$$

$$= \frac{1 \pm \sqrt{1 + 4}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

$So\ we\ get$:

$$\varphi = \frac{1 + \sqrt{5}}{2} = \frac{1 + 2.236}{2} = \frac{3.236}{2} \approx 1.618$$

$$\psi = \frac{1 - \sqrt{5}}{2} = \frac{1 - 2.236}{2} = \frac{-1.236}{2} \approx -0.618$$

$$F(n) = A\varphi^n + B\psi^n$$

$\boldsymbol{As, \psi^n =}$

$\boldsymbol{(-0.618)^2 = 0.382}$

$\boldsymbol{(-0.618)^3 = -0.236}$

$\boldsymbol{(-0.618)^4 = 0.145}$

$\boldsymbol{(-0.618)^5 = -0.090}$

$\boldsymbol{\ldots\ldots}$

$\boldsymbol{Hence\ we\ can\ tell:\ \psi^n \to 0}$.

$\boldsymbol{Where, \varphi^n = (1.618)^n \to \infty}$

Term	*Value*	*Behavior*
$\boldsymbol{\varphi}$	1.618	*grows exponentially*
$\boldsymbol{\psi}$	-0.618	*shrinks to* 0

$\boldsymbol{Therefore\ we\ can\ say: \Theta(\varphi^n), where\ `\varphi = 1.618`}$.

$\boldsymbol{And, its\ precise\ and\ Actual.}$

$\boldsymbol{Also, in\ Time\ Complexity\ we\ have\ seen}$:

$\boldsymbol{T(n) = T(n-1) + T(n-2) + O(1)\ or, T(n-1) + T(n-2) + 1}$
$\boldsymbol{\Rightarrow O(2^n)}$ [$\boldsymbol{This\ is\ also\ not\ precise.}$]

$\boldsymbol{Lets\ say,}$

$\boldsymbol{T(n) = x^n}$

$\boldsymbol{Then\ again\ , we\ get}$:

$\boldsymbol{\Theta(\varphi^n)\ , where\ `\varphi\ = 1.618\ `}$

$\boldsymbol{That\ is\ Growth\ Rate\ of\ Output = \ Time\ Complexity\ here\ , but\ its\ a\ coincedence\ here\ , i.e. not\ always\ True.}$